Ephemeral Interfaces: Design Patterns for Task-Scoped Generative UI Components

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Capstone Project

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I would like to thank everyone who has helped guide me through the process of creating this thesis. It was a personal idea I really wanted to explore and felt empowered to try this with all the positive feedback I got from friends and colleagues.

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Let the capstone begin.

Abstract

Generative UI has become a broader topic in software design because it promises interfaces that adapt to user goals instead of remaining static. Yet this promise collides with modern software use, where users still expect stable experiences, predictable controls, and durable mental models. This thesis investigates whether ephemeral, task-scoped generative UI components can offer a bounded alternative to full interface regeneration by appearing only when locally useful and disappearing without destabilising the host application.

The thesis develops and tests a framework for ephemeral support through a web-based research artifact. The main individual contributions are the framework operationalisation, a constrained component-specification model for temporary support, a validation pipeline for generated interface specifications, a bounded ephemeral overlay approach that preserves interface continuity, and an interactive survey artifact for empirical evaluation. The artifact was tested in a within-subjects user study comparing a baseline interface with an ephemeral condition on a bounded slide-reordering task.

The results show a mixed outcome. The ephemeral condition showed a descriptively higher task completion rate and significantly higher helpfulness ratings, while perceived control remained stable, but it did not improve completion time or overall preference and significantly increased interaction count and perceived intrusiveness. These findings suggest that ephemeral generative support is most credible not as a general replacement for existing interfaces, but as a bounded design strategy for bounded, low-consequence subtasks within broader workflows, especially where lightweight contextual guidance may be useful and interruption costs remain manageable. The thesis therefore contributes a framework, design guidelines, and an empirical basis for assessing when temporary generative assistance can add value without disrupting the user experience.

Keywords: jgenerative AI, Design, UI/UX, Web apps, Ephemeral interfaces, Human-computer interaction, Productivity software, Design science research

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1 Introduction

1.1 Problem and Motivation

Large language models are not only limited by text generation, but are often used to make user interfaces that can change based on the users' needs. This shift has expanded the structure of an application should be designed to respond to the needs of the user. But the promise of AI-generated interfaces that adapt over time creates a conflict of interest. On one hand, static interfaces provide users a constrained and predictable user journey for them to follow to complete a certain task but may not match the needs of diverse real world problems. On the other hand, fully generative approaches that replace the entire interface risk losing the benefits of a predictable and continuous interface that current day software provides.

This tension affects my own motivation to work on this project. In my experience working as a software engineer on production applications, generative AI is often introduced through chat-based interfaces used in tandem with the main product experience. Some of these systems have also begun to incorporate generative UI to present information more visually. But there is a clear gap in how to add adaptive support to current systems without using a chat-based workflow.

Furthermore, these problems are especially acute in productivity software, where users build durable mental models of how their tools behave. When an interface is regenerated anew in response to each new request, orientation is lost so users are required to relearn the interface, how to use it, and where key controls are located. The design challenge, therefore, is not simply whether AI can generate useful interfaces, but how generative capability can be introduced into an existing application without eroding the stability that makes it usable in the first place.

A promising middle ground lies in *ephemerality*: rather than replacing an interface entirely, the system could surface temporary, task-scoped components that appear only when contextually justified and disappear once their purpose has been served. Such components would augment the host application at the point of need while leaving the surrounding interface intact. Although this idea draws on established principles from both generative UI research and

interaction design theory, a coherent framework that specifies when ephemeral components should appear, how their scope should be bounded, and how they should be withdrawn without disrupting the user's workflow has not been articulated in prior design work.

1.2 Literature Review

While LLMs have been primarily used to generate content through text, they are becoming more prevalent in the generation of user interfaces (UI). It is a way to move beyond static interfaces by generating structures based on an underlying set of primitives tailored to the users' goals. Prior work suggests that when working with LLMs, users prefer to express themselves through UI instead of a plain chat. This is because the system can present its functionality properly instead of forcing it through text (Ahmed & Imran, 2025; Leviathan et al., 2025; Tang et al., 2025). However, most of the evidence comes from fully generated, end-to-end user interfaces, which differs from the design challenge addressed in this thesis, which is to augment existing applications where users expect a certain degree of stability and predictability in their tools. This raises the question of how an underlying set of core primitive UI components can be integrated into already established workflows instead of replacing the entire state of the interface with AI.

The problem with generative UI interfaces is that while they can increase the relevance of the task at hand, they can threaten predictability and user control. In real-world domains such as banking, their rigid interfaces were built on a set of predefined rules that fail to meet the ever-changing needs of their users across services (Tambi, 2020). Approaches that dynamically generate interfaces can address the problem of not tailoring just to the content needed to serve the user, but the actual structure of the interaction over time. Yet, these new systems raise an unresolved question: when generative UI is beneficial, how can it be introduced to users without it corroding the user's mental model of how the application should work? This means that the design guidelines for generative UI must balance adaptivity and consistency (Bieniek et al., 2024)

To manage this balance instead of having the entire interface generated by an LLM we restrict it to an ephemeral boundary. Thus, task-scoped components will only appear and disap-

pear when needed. Past work on ephemeral interfaces argues that the permanence in ubiquitous environments can contribute to cognitive overload as there are too many controls and artifacts present (Döring et al., 2013). Conversely, ephemerality can reduce the persistent clutter of artifacts and help with better interactions for the user within a specific time frame and context. While this research was grounded in physical interaction design, it provides principles that can be transferred to digital systems. These ephemeral elements have to be easily discoverable, dismissible, and disappear without affecting the surrounding environment. Furthermore, it's a great use case for LLMs as the system could generate UI contextually in almost real time. (Bieniek et al., 2024). However, it must be kept in mind how the component lifecycle should be carefully designed to prevent an unstable user experience.

Ephemeral generative components have been implemented in past research without a full interface replacement, but with a temporary UI as an intermediate layer for exploration and understanding. For example, research was done in data analysis where they had to tweak their results from their machine learning models constantly through their code. To improve observability and comprehension of the LLM generated code an ephemeral interface was used instead of manually tweaking the values (Cheng et al., 2024). This shows how ephemeral UI can help users steer their outcome and iteratively refine stochastic tasks for exploring alternatives before taking action. However, this application of ephemeral UI was done in a narrow domain and intent. This leaves the question of how ephemeral patterns can generalise to everyday productivity applications where tasks are diverse. If it's not implemented properly, it could lead to more interruptions, context switching, or inconsistency of a familiar workflow.

Based on reviewed literature, it establishes three incomplete variables to the capstone. Firstly, a fully generative UI can improve user experience of interacting with applications compared to only chat-based interactions (Leviathan et al., 2025; Tang et al., 2025). Secondly, using dynamic interface generation in specific domains with ever evolving problems and rule based logic that require tailoring (Tambi, 2020). Thirdly, research on ephemerality shows that cognitive load can be reduced by decreasing the amount of permanent UI elements for the user. Nonetheless a gap remains at actually providing guidance on how to design these ephemeral, task-scoped generative UI components that can improve existing applications without affecting

negatively the user experience. Prior work does not standardise reproducible design patterns of when generative components should appear, their bounded scope with an exit, and how they should preserve a predictable user interface. This capstone addresses those gaps by standardising ephemerality into a framework of component patterns and rules to integrate into everyday applications.

1.3 Research Question and Contribution

The gap identified in the literature motivates the following research question:

> *How can ephemeral, task-scoped generative UI components be designed to improve existing applications without disrupting the user experience?*

The capstone address this question in several contributions. Firstly it aggregates a **conceptual framework** that defines a taxonomy of ephemeral components, a component lifecycle, and a set of design principles to constrain these temporary generative UI components in stable interfaces. Secondly, it **operationalises that framework** through a constrained component-specification model that defines what kinds of temporary support may be generated and how they may be structured. Third, it implements a **validation pipeline and bounded overlay approach** that checks generated specifications against structural and scenario constraints before rendering them as temporary, dismissible interface layers that preserve continuity with the host application. Fourth, it builds an **interactive survey artifact**: an experimental website that implements both a conventional baseline interface and an ephemeral condition governed by the framework's rules, with event logging and questionnaire capture built into the same system. Fifth, it evaluates the research through a **comparative within-subjects user study** that measures task completion, efficiency, interaction effort, perceived helpfulness, intrusiveness, and user control across both conditions. These contributions go beyond existing literature by offering a empirical evidence of whether ephemeral UI can improve productivity scenarios

1.4 Thesis Outline

The remainder of this thesis is organised as follows. Section 2 presents the research methodology, including the conceptual framework, the design of the artifact, and the evaluation plan. Section 3 describes the technical implementation of the experimental website and its support-generation pipeline. Section 4 reports the results of the user study across all behavioural and self-reported measures. Section 5 interprets the findings in relation to prior work and discusses the limitations and robustness of the evaluation. Section 6 outlines directions for future work, and Section 7 concludes.

2 Methodology

2.1 Research Design

Design Science Research (DSR) (Delport et al., 2024) was used as its primary goal was not only to show an existing phenomena but to create and evaluate an artifact capable of replicating the user identifying the design problem. The problem established in the literature was that current UI interfaces are either constrained completely to a static interface or the entire experience is dynamically generated resulting in unpredictably. Therefore, this research proposes the conceptual framework for ephemeral, task-scoped generative UI components to augment existing applications without disrupting the user experience. The methodological contribution lies in developing a framework that translates ideas from previous work on ephemerality and generative UI to guide better design decisions in a broader productivity context.

The study proceeds in three connected stages. Firstly, a framework based on existing literature on defining a taxonomy, lifecycle, and design principles for ephemeral interface components. Secondly, the framework is put into practice through a web-based research instrument artifact that allows these proposed ideas to be conducted in a controlled environment. Third, evaluation is conducted through the comparative user study by comparing a traditional interface with an ephemeral one on a single bounded task. By using the framework we can construct the artifact that will provide the basis for whether ephemeral UI can improve task performance and user experience.

2.2 Conceptual Framework

This study develops a conceptual framework for ephemeral, task-scoped generative UI components to guide both artifact design and evaluation. In one sentence, the framework specifies *which temporary support roles may appear*, *how those supports are triggered and withdrawn*, and *which constraints keep them useful without destabilising the host interface*. Positioning the framework here in the methodology is important because it defines the concepts that the artifact operationalises and the criteria against which the two interface conditions are later compared.

The first element is a taxonomy of component roles. Rather than classifying components by visual form, such as whether they appear as a panel, popover, or inline element, the taxonomy classifies them by the function they serve within the workflow. Three roles are distinguished. *Interpretive support* helps users understand context, available options, outputs, or likely consequences before acting. *Refinement support* helps users steer or configure an intended action before commitment, for example by comparing alternatives or adjusting parameters. *Task-execution support* helps users complete a bounded subtask inside the surrounding application without replacing the broader workflow. Framing these roles as a taxonomy is methodologically useful because it makes explicit what kind of assistance is being introduced in each scenario and avoids treating ephemeral behaviour as a single undifferentiated interface style.

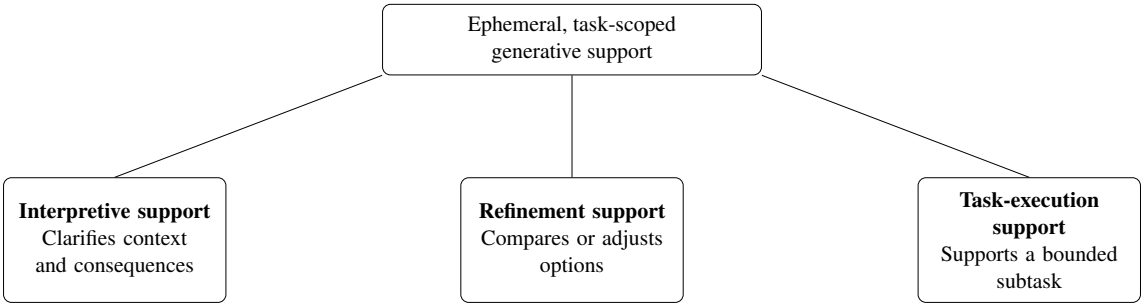


Figure 1: Taxonomy of ephemeral, task-scoped generative support roles.

The second element is a lifecycle describing how an ephemeral component enters and leaves use. Alternative stage patterns can be found in the literature, ranging from the minimal logic of temporary appearance and disappearance (Döring et al., 2013) to more scaffolded models in which users inspect and steer an intermediate layer before committing to an outcome (Cheng et al., 2024). For this study, the most appropriate synthesis is a four-stage lifecycle of *trigger*, *instantiation*, *interaction*, and *dismissal*. The component is triggered when a clear user goal or contextual need is detected that is not adequately supported by the persistent interface alone. That trigger may be explicit, such as a user request for help, or inferred from local behavioural and contextual signals, but only when the resulting support remains low-risk, relevant, and easy to dismiss. The component is then instantiated as a temporary interface element embedded within the existing application rather than replacing the surrounding interface. During interaction, users inspect, steer, or complete the immediate subtask while remaining oriented within the primary workflow. Finally, the component is dismissed once the task is

completed, abandoned, or no longer contextually relevant. This lifecycle was selected because it retains the conceptual clarity of ephemerality while remaining concrete enough to implement as a state-based interaction model in the artifact.

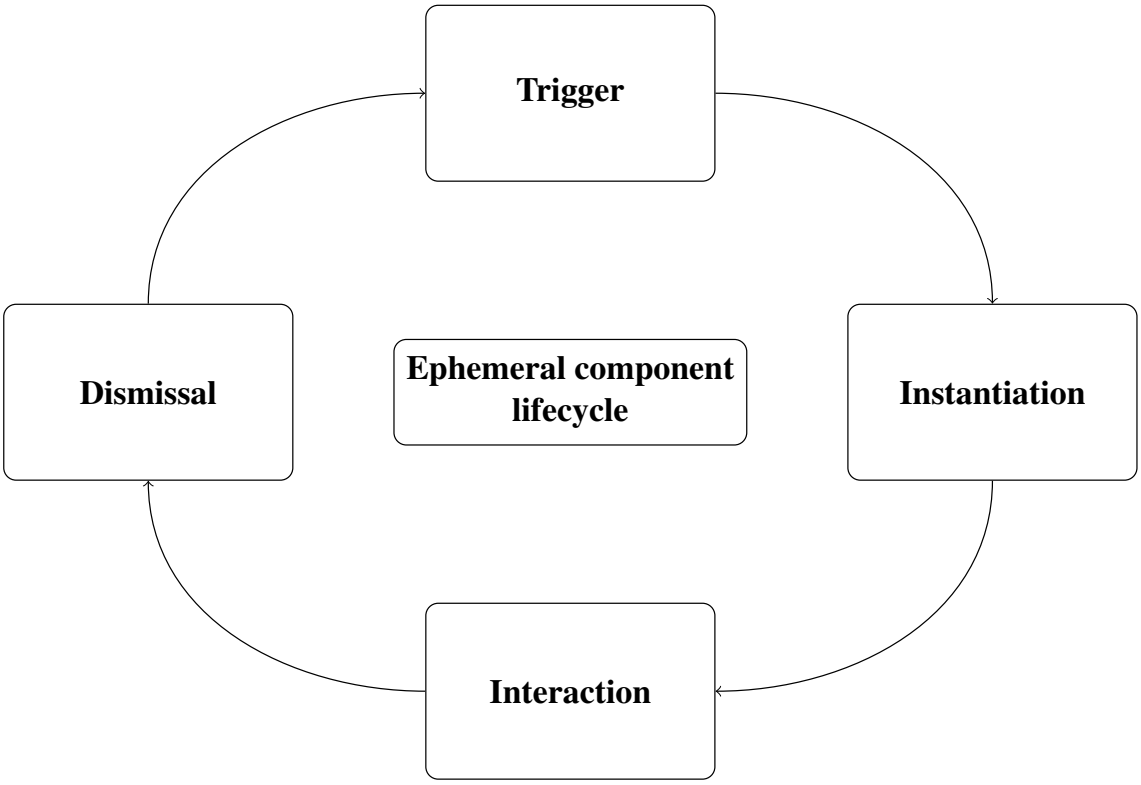


Figure 2: Lifecycle of an ephemeral, task-scoped generative UI component: trigger, instantiation, interaction, and dismissal.

The third element are design principles that constrain how ephemeral components should be introduced into a stable interface. Five principles guide the framework. **Contextual relevance** which determines the reason for when support should appear and serve a particular task. **Bounded scope** defines that the component should only remain present for the immediate task rather than the overall workflow. **Interface continuity** is where the component should preserve its surrounding elements so it feels part of the experience instead of redirecting the user to an unrelated experience. **User control** means that users should be able to inspect, refine, confirm, or dismiss the support rather than being forced to take a decision by the system. **Graceful disappearance** is used so that the component should disappear gracefully once it is not relevant to the immediate task at hand. If the ephemeral support is inferred rather than invoked explicitly then the defined principles constrain inferred assistance so consequential actions such as performing a destructive action to delete data is done without explicit user confirmation. Thus,

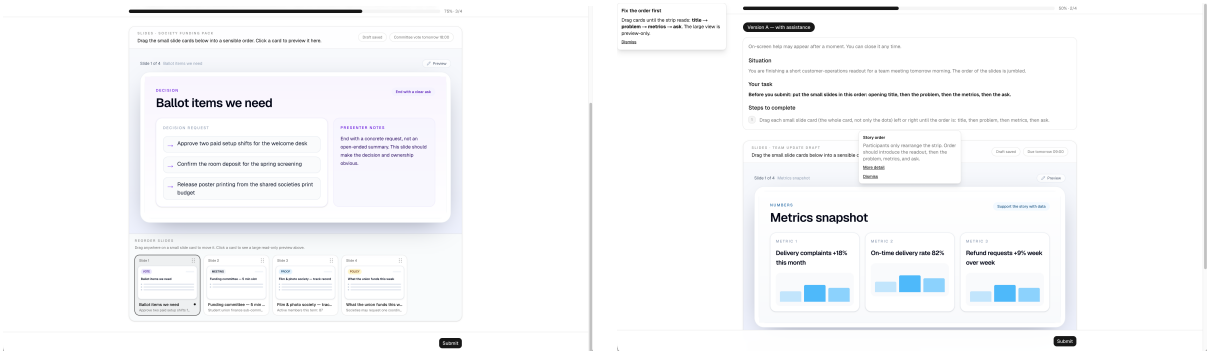
these principles communicate the idea of ephemerality into practical design criteria that can be used consistently during the artifact development.

2.3 Artifact Development

The artifact developed in this project is a website used as a research instrument for evaluating the proposed framework in a controlled but realistic interaction setting. Rather than building a fully autonomous or fully generative application, the website is designed to preserve a recognisable baseline interface while allowing ephemeral, task-scoped components to be introduced at selected moments. This makes it possible to test whether the framework can augment the existing user journey without replacing the primary workflow altogether.

To operationalise the framework, the website will implement two interface conditions: an original condition using a well-known, conventional interface and an ephemeral condition in which temporary, task-scoped components are introduced according to the taxonomy, lifecycle, and design principles defined above. Both conditions use the same underlying slide-deck interface, task mechanics, and submission logic. The original condition presents the task without temporary support, whereas the ephemeral condition will surface temporary interface elements only when additional guidance, refinement, or task support is needed. In this way, the comparison isolates the effect of ephemeral augmentation against the non-ephemeral alternative.

The website presents one scenario that represents a bounded productivity task within the application: a short presentation-refinement exercise foregrounding *refinement support*. Participants receive a small slide deck whose order has been intentionally scrambled and must restore it to a coherent narrative sequence before submitting. The interaction is deliberately narrow: the main slide canvas functions as a large read-only preview, while the participant performs the task by reordering the slide thumbnails. The lifecycle of trigger, instantiation, interaction, and dismissal is implemented as the common interaction logic for both interface conditions. In the ephemeral condition, temporary support may propose order cues, comparisons, or lightweight explanations of narrative flow, but it does not edit slide content or take over the task. Success is therefore defined as submitting the deck in the predefined canonical order. Figure 3 shows the implemented study interface in both conditions.



(a) Baseline condition without temporary support.

(b) Ephemeral condition with temporary support visible.

Figure 3: Screenshots of the implemented experiment interface. In both conditions, participants complete the same slide-reordering task: the large slide canvas is a read-only preview, while the actual task action is performed through the thumbnail strip. The ephemeral condition preserves the same underlying task structure but adds temporary, dismissible support overlays.

The ephemeral condition relies on a constrained support-generation pipeline rather than unrestricted interface generation. When support is requested or triggered, the system produces a structured interface specification from the current task context, validates it against schema and safety constraints, and either renders it as a temporary overlay or falls back to a deterministic alternative. The technical details of this pipeline, including the specification model, the validation stages, and the rendering layer, are described in Section 3.

Although the ephemeral condition is allowed to be visually and structurally expressive, that expressiveness remains bounded by the framework. Temporary support may add task-relevant interface elements, annotate or highlight existing content, expose comparison views, or locally reconfigure a small region of the page. However, it does not replace the full application, arbitrarily reorganise the entire interface, or perform consequential actions without confirmation. In practice, the role category, lifecycle, task goals, and success criteria are fixed in advance, while local details such as wording, defaults, and the exact form of low-risk support may vary in response to immediate context. The ephemeral version is therefore not intended to be wholly different for every participant. Instead, it is **constrained adaptation**: users encounter the same scenario structures and the same framework rules, but the temporary support may be parameterised slightly differently within those limits. This preserves analytical robustness while still allowing the artifact to reflect the broader motivation of adaptive, goal-relevant ephemeral interfaces.

2.4 Evaluation Design

2.4.1 Experimental Design

The framework will be evaluated through a comparative within-subjects user study using the website artifact described above. Each participant will complete two trials in total: the same slide-refinement scenario once as Version A and once as Version B. The presentation order of those versions is fixed, with Version A always shown first and Version B second, but the mapping between version label and experimental condition is randomised per participant so that one participant may receive the baseline first while another receives the ephemeral condition first under the alternative label. A within-subjects design is appropriate because it allows each participant to serve as their own control, thereby reducing the influence of individual differences in participants' digital literacy, task strategy, and previously acquired familiarity with similar interfaces. Having version labels preserves a consistent study flow while still varying which version corresponds to assistance across participants.

2.4.2 Participants

Participants will be recruited as regular users of web-based applications who can complete common digital tasks in a desktop browser environment. Recruitment will follow a convenience sampling approach appropriate to the scope of the capstone project. Inclusion criteria will focus on participants who are able to use standard web interfaces independently, while any exclusion criteria will be limited to factors that would prevent valid completion of the study tasks. The final sample size will depend on recruitment feasibility, but the study is designed to support repeated-measures comparisons between the two interface conditions.

2.4.3 Materials and Procedure

The main experimental artifact is the website. Participants complete the same scenario but with different context twice. The task is that they must rearrange a slide deck for a given correct sequence. The structure, goals, and success criteria remain constant for all users so that the study remains comparable across sessions. The ephemeral condition varies in how it's pre-

sented across sessions so that the assistance from the LLM can respond to given context while constrained to the predefined design rules from the framework. Thus, this study does not test unlimited personalisation or an unrestricted interface takeover. The main test is how bounded interfaces with temporary support could add, highlight, compare, or rearrange elements while preserving interface continuity and user control.

The procedure will be fully web-based. After opening the study website, participants will read the study information, provide consent, and complete a short background questionnaire about their comfort with websites and familiarity with AI-assisted tools. The website will then present two task trials in a fixed sequence of Version A followed by Version B, representing the same scenario under the two experimental conditions, with the condition-to-version mapping randomised per participant. Before each trial, participants will receive a short task prompt that labels the interface version. During each trial, the website will automatically log behavioural data. After both trials are done, participants will fill out a final comparative questionnaire about which condition they liked better, with a clear reminder of which version helped them. This fully instrumented procedure maintains the study's structure while facilitating the collection of all behavioural and self-reported data within a unified environment.

2.4.4 Measures

The evaluation combines both self-reported and behavioural measure to encapsulate task performance and perceived experience from the user. The Behavioural measures are derived from the interaction logs logged from the website. Subjective measures are collected through post-trial questionnaires. What is being tested is not only that ephemeral interfaces are preferred, but that they can actually improve how well a person can complete a task without external disruption. To assess that claim, efficiency measures task completion time, effectiveness through task completion rate, friction through interaction count. While user experience will be measured through ratings of intrusiveness, perceived control, helpfulness, and overall preference. Furthermore, ephemeral interactions like the time when ephemeral support is triggered, requested, shown, used, dismissed, ignored, or expanded for further detail is logged. It allows the study to gain further insight into how participants engaged with the generated assistance.

Table 1: Dependent measures and collection methods.

Measure	Type	Collection	Analysis
Task completion rate	Binary	Event logging	McNemar's exact test
Task completion time	Continuous	Timestamps	Wilcoxon signed-rank test
Interaction count	Discrete	Event logging	Wilcoxon signed-rank test
Intrusiveness	Ordinal (1–7)	Post-trial Likert item	Wilcoxon signed-rank test
Perceived control	Ordinal (1–7)	Post-trial Likert item	Wilcoxon signed-rank test
Helpfulness	Ordinal (1–7)	Post-trial Likert item	Wilcoxon signed-rank test
Preference	Nominal	Final forced choice	Binomial test

2.4.5 Data Collection

Data collection will combine instrumented event logging with questionnaire responses. For each trial, the website will record task start and end times, completion outcomes, interaction counts, submitted task state, and the active interface condition. In the ephemeral condition, logs from ephemeral interactions will capture support trigger events, explicit requests, support shown events, dismissals, usage confirmations, ignored support, detail expansions, and the generated support outputs themselves. Post-trial survey items will capture perceived intrusiveness, perceived control, and helpfulness, while the final questionnaire will capture overall preference between conditions. To link behavioural and survey data, each session is associated with a cookie-linked participant identifier that is stored consistently across the website logs and questionnaire responses. No names or email addresses are collected; the study stores only age range, occupation, familiarity ratings, trial logs, and questionnaire responses.

2.4.6 Analysis Plan

The primary analysis will compare the original and ephemeral conditions on each dependent measure using paired statistical tests selected for the data type, scale of measurement, and expected sample size.

Task completion rate is a paired binary outcome: each participant either completes the task correctly or does not, under each condition. This will be analysed using McNemar's exact test, which evaluates whether the proportion of participants who change completion status between conditions differs from what would be expected by chance. Completion proportions for each

condition will be reported alongside the test result.

Task completion time and interaction count are continuous and discrete measures, respectively, that are paired across conditions. The Wilcoxon signed-rank test, which does not assume normality of the paired differences, will be used to analyse both measures because the expected sample size is small and time data is usually right-skewed. Medians and interquartile ranges will be reported for each condition.

Post-trial Likert ratings for intrusiveness, perceived control, and helpfulness are ordinal measures on a scale of one to seven. Each rating will be analysed using the Wilcoxon signed-rank test, which is appropriate for paired ordinal data. Medians will be reported rather than means to respect the ordinal nature of the scale.

Overall preference is a forced-choice measure in which each participant selects which condition they preferred. This will be analysed using a binomial test against the null hypothesis that preference is split equally between conditions. The number and percentage of people who chose each condition will be reported.

A descriptive analysis of ephemeral interaction logs will show the percentage of participants who triggered, displayed, used, ignored, missed, or added to support. This analysis aids in determining whether a null or positive effect indicates the quality of the support or a lack of engagement. In subsequent analysis, the relationship between increased efficiency or perceived helpfulness and minimal interruption and perceived control was investigated. The framework allows the ephemeral layer to change, so any changes to the rules are seen as part of the design rather than random changes.

3 Implementation

The previous section defines the framework and theory of how the artifact will be used to evaluate the comparative study. The following section provides the technical breakdown of how the artifact will actually be made. The goal is to explain the technical design legible enough for the reader to assess whether the artifact can use the framework and instrumentation sufficiently for the evaluation.

3.1 Architecture Overview

The artifact is a web app built with Next.js and TypeScript (Vercel, 2026b). It has API routes for managing participants, trials, event logging, and metadata for ephemeral UI. Persistent data is stored in a PostgreSQL database. Then, the ephemeral UI engine uses the Google Gemini API through the Vercel AI SDK to ensure structured data outputs from the LLM based on the current task state (Google, 2026; Vercel, 2026a).

The choices made are based on the scope of only the research done in the capstone rather than made for a real production deployment. Next.js was used as it's trivial to spin up a colocated backend and frontend for fast deployment, which simplifies the instrumentation of generating the ephemeral UI and handling user interactions. TypeScript and Zod were used for compile and runtime safety of the generated objects from the generative AI pipeline to make sure that all outputs are validated with respect to a specification. Therefore, we could ensure a level of deterministic behaviour in the algorithm. PostgreSQL offered the relational integrity for the structured experimental data. The Vercel AI SDK (Vercel, 2026a) abstracts all the LLM's provider specific details when trying to have structured outputs, resulting in the deterministic JSON output that can be used to render the UI on the web app.

3.2 Participant Flow and Session Management

The artifact described in the methodology has a linear workflow that the participant must follow to finish it. When they first open the website information about the capstone is provided with a consent form. Once consent is given, a participant record is created in the database and sets

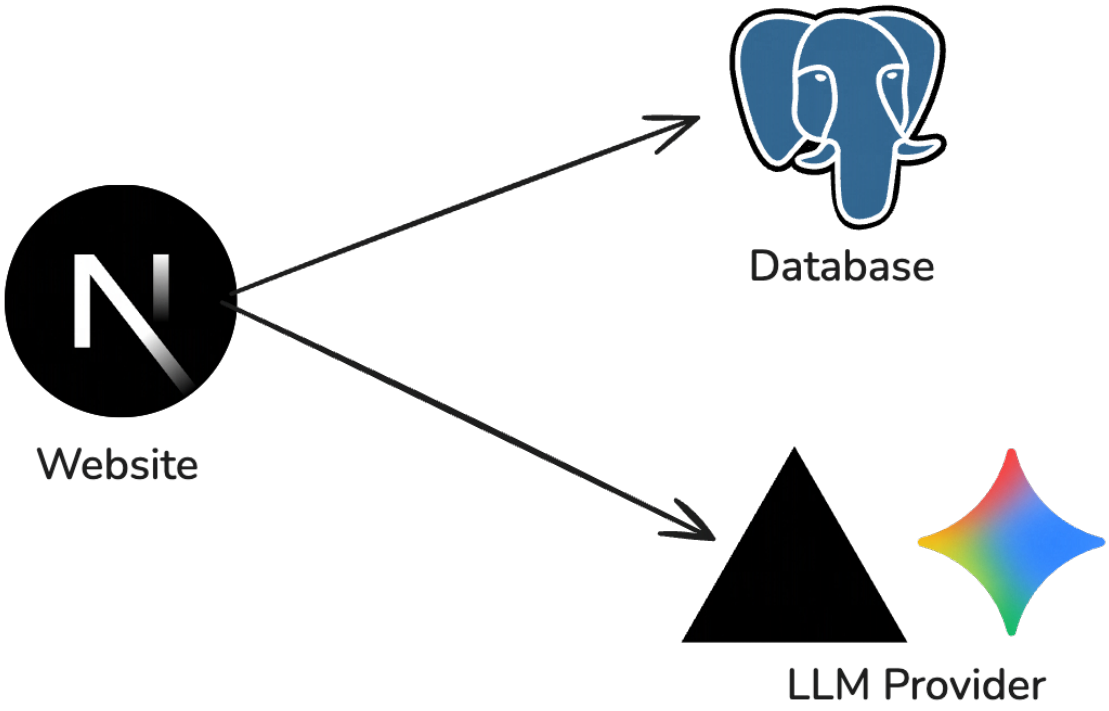


Figure 4: High-level system architecture of the experimental artifact.

a session cookie that associates all subsequent requests with it for an identifiable session. No traditional authentication is used to preserve the anonymity of the participant.

Once the participant has consented, they follow a short background questionnaire including their age, occupation, familiarity with web applications, and familiarity with AI-assisted tools. After submitting their response, it creates the first trial row in the database. From here on, a backend process determines the current state the user is in. These possible states contain a fixed sequence of *background*, *study*, *post-trial*, *questionnaire*, *final questionnaire*, *complete*. Each route in the app is mapped to every state so reloading the page persists the current state the user is in.

Trial rows are created incrementally rather than all at once. The first trial is created when the participant acknowledges the instructions. Each subsequent trial is created only after the previous trial has been completed and its post-trial questionnaire has been submitted. This ensures that the participant cannot skip ahead or access later trials before finishing earlier ones.

To make sure that results are counterbalanced between participants, when a participant record is created a boolean flag randomly determines whether the baseline is Version A or Version B. Then the trial assigns conditions the version labels from this boolean flag so that

one participant could get the baseline version first or the ephemeral version under the label "Version A." The content variant for each condition is determined by hashing the participant and scenario identifiers, resulting in distinct slide decks for the baseline and ephemeral trials.

3.3 Data Model

The database schema consists of five tables designed to capture the behavioural and self-reported data required by the evaluation measures defined in the methodology. Figure 5 shows the entity–relationship diagram (ERD): each participant may complete several trials; each trial may emit many logged events and many stored ephemeral support generations; questionnaire responses belong to a participant, and may optionally reference a trial (post-trial items) or omit a trial key (final questionnaire). Table 2 summarises what each table contributes to the evaluation.

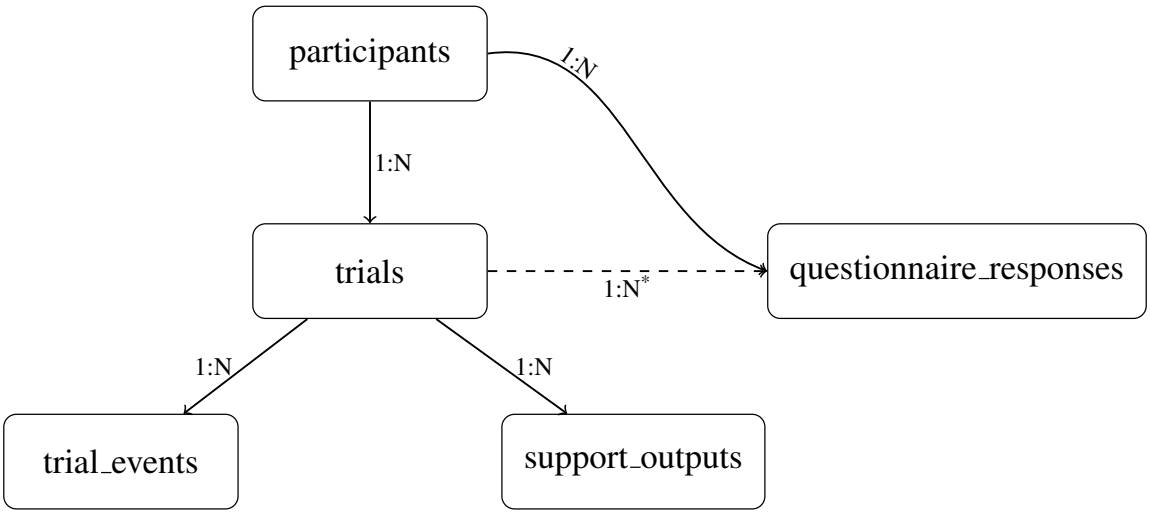


Figure 5: Entity–relationship diagram of the study database.

The schema preserves referential integrity and supports the study queries needed for export and analysis.

3.4 Task Scenario Implementation

The study evaluates a single scenario where the user reorders slides based from the methodology. The implementation uses a registry of scenarios where each scenario has its own taxonomy, correct answer, allowlist of ephemeral targets, task context, and function for initial task

Table 2: Summary of each table's role relative to the evaluation measures.

Table	Role in the study
`participants`	Consent and background questionnaire; counterbalancing flag (`baseline_is_version_a`); links all rows for a session.
`trials`	One row per task trial: condition, scenario, timing, correctness, submitted answer, interaction count—supports performance and efficiency measures.
`trial_events`	Typed events with JSON payloads—supports interaction counts and ephemeral engagement (e.g. support shown, dismissed).
`questionnaire_responses`	Post-trial Likert items and final preference; `trial_id` set for post-trial rows, null for final rows—supports subjective measures.
`support_outputs`	Ephemeral condition only: model input state and output spec per generation—supports analysis of support quality and fallback.

state. Thus, the design allows the ease of adding new scenarios when needed without changing the rest of the app.

Furthermore, the scenario provides participants a minimal four slide deck intentionally scrambled. The main content area shows a read-only preview for every slide, while the actual interaction happens underneath the preview of the slide through a horizontal strip of draggable cards. Drag and drop was implemented using the `@dnd-kit` npm package (Claudéric Demers, 2026), which provides an abstraction over pointer-based reordering. The main goal of the scenario is for the participants to reorder the slides and submit what they think is correct relative to the context of the scenario's question.

A submission from the participant is valid if the order they chose is equal to exact predefined order from the given variant. If they are not exactly the same, it is not correct. Variant A is a pitch deck with the slides of title, problem, metrics, and ask. While, variant B is a committee funding slide deck with slides of meeting, policy, proof, ballot items. Both variant A and B use a deterministic hash of the participant identifier to the baseline condition, with the opposite variant for the ephemeral condition. So, each person will see the two scenarios in a random order, which will keep them from getting used to the task content and keep the task structure the same.

Both variations share the exact same interaction mechanics and submission logic. The main

difference between either are whether ephemeral support is active or not. Thus, it isolates the effect of using ephemeral UI from other possible variables.

3.5 Ephemeral Support System

The generative AI pipeline that makes the temporary overlays is the main technical contribution of the artifact, by using the framework's taxonomy, lifecycle, and design principles. The following subsection defines the three layers of how the system works: the specification model of how it can be generated, the pipeline that creates and validates specifications, and the rendering layer that displays them.

3.5.1 Component Specification Model

Ephemeral support is shown as a typed tree structure as `EphemeralSpec`. Each element has a version number, a root node, and metadata that tells you if the overlay can be removed or not. All nodes are recursive, so each node has a type, property bag validated against a typed schema, and optional children. Additionally, the tree is constrained to a maximum depth of four levels and six children per node. The decision to use JSON as the intermediate structure for these specifications was inspired by JSON Render, which similarly demonstrates how interface descriptions can be expressed declaratively and then rendered from structured data (Jason Lengstorf, 2026).

The component catalog defines seventeen component types, organised into five functional categories. Table 3 lists the types and their roles. Visual cue components draw attention to specific interface elements without adding content. Anchored annotation components attach textual or rich content near a target element. Relational components connect or compare two or more targets to support reasoning across elements. Layout containers position children within the viewport or relative to a target. Rich content components render sanitised HTML and inline SVG for more expressive explanations such as flow diagrams or annotated illustrations.

Each component type has a Zod schema that validates its properties at runtime. These schemas limit variables like the maximum length of strings, the range of numbers, and the fields that must be filled out. For example, an `InspectPanel` ephemeral component that explains to

Table 3: Ephemeral component catalog. Each type is validated against a per-component Zod schema that enforces field types, string lengths, and structural constraints.

Category	Component type	Role
Visual cues	`HighlightRing`	Amber ring around a target element
	`PulseRing`	Animated ring with configurable duration
	`ArrowCue`	Arrow pointing at a target
	`FocusMask`	Dims everything except the target
Anchored annotations	`AnchoredTooltip`	Plain-text tooltip near a target
	`HintStack`	Ordered list of short hints near a target
	`InspectPanel`	Expandable panel with summary and optional detail lines
	`ConsequenceNote`	Single italic consequence line near a target
	`SideNote`	Margin annotation beside a target
Relational	`ConnectorLine`	Dashed line between two targets
	`ComparisonStrip`	Short contrast text positioned under two targets
	`StepRail`	Numbered callouts across two to six targets
Layout containers	`Stack`	Vertical sequence container
	`ViewportPanel`	Absolutely positioned panel in viewport percentages
	`TargetOffsetPanel`	Panel positioned relative to a target with pixel offsets
Rich content	`FlowHtml`	Sanitised HTML block (only inside `ViewportPanel`)
	`AnchoredHtml`	Sanitised HTML anchored near a target

a participant why the order of the slides in a pitch deck is wrong still needs to include a target identifier, a title of no more than 120 characters, a summary of no more than 280 characters, and up to four detail lines of 200 characters each. These limits stop the model from making components that are not design and situation relevant.

The catalogue is versioned so that you can keep track of changes to the types of components that are available and their restrictions over time.

3.5.2 Generation and Validation Pipeline

When ephemeral UI is needed during a trial, the system runs the pipeline to make a specification validating it through the catalogue and scenario constraints. It either accepts it or falls back to deterministic alternative. Figure 6 illustrates this pipeline.

To generate the UI components dynamically prompt engineering is required. Two prompts were used in the system. Firstly, the system prompt includes the specification schema, component types with their property schemas, target identifier states, taxonomy, and design rules.

Then, the user prompt contains the information of the participant. That includes task state, order of the slides with metadata, review goals, and an optional snapshot of participant interaction. The snapshot is the participant's progress the moment ephemeral support was requested. It allows the model to tailor the output of what the user has already done instead of assuming it's the user's first interaction with the interface.

Thereafter, Vercel AI SDK's structured output sends these prompts to the Google Gemini API (Google, 2026; Vercel, 2026a), following the Zod schema that was given. The output from the model is then processed several validation stages:

1. **Envelope parsing.** Validates structure to make sure it has a valid version number, root node, and metadata.

2. **Node tree parsing.** The root and its children are checked recursively to make sure they are the right shape, have the right number of children, and with reasonable depth.

3. **Per-component property validation.** Each node's properties are validated against its Zod schema.

4. **Target allowlisting.** Every target identifier is checked against the list of targets that are allowed. If an interface element isn't listed in the scenario registry, it will fail.

5. **Structural container rules.** Layout containers are verified to make sure they have at least one child. This is true for `Stack`, `ViewportPanel`, and `TargetOffsetPanel`.

6. **Ancestry constraints.** Only `FlowHtml` nodes are allowed in a `ViewportPanel` subtree. This stops rich HTML from being able to float around anywhere in the interface.

7. **HTML sanitisation.** DOMPurify cleans up any HTML in `FlowHtml`, `AnchoredHtml`, or `SideNote` nodes by letting you specify which tags and attributes to use. This stops harmful JavaScript code from being injected while letting semantic HTML and inline SVG for diagrams.

If any stage rejects the output, the system falls back to a deterministic, hardcoded specification for the current scenario. Fallback specifications are defined per scenario and per variant,

ensuring that every participant in the ephemeral condition receives some form of support regardless of model behaviour. The fallback specifications use the same component types and target identifiers as model-generated output, so they appear visually consistent with the rest of the ephemeral experience.

Whether the specification was model-generated or a fallback, the full input state, model name, and output are persisted to the `support_outputs` table. This logging enables post-study analysis of generation quality, fallback frequency, and the relationship between input context and output content.

To verify the implementation end-to-end outside the aggregate study analysis, a local support-generation workflow was also conducted on 2026-04-04. Three consecutive hesitation-triggered requests for the slide-reordering scenario completed successfully and were persisted as support-output records without fallback. A non-trivial multi-component overlay was generated, validated, saved to PostgreSQL, and rendered successfully. Exact trace details are provided in Appendix A.3.

3.5.3 Rendering Layer

Accepted specifications are rendered as a temporary overlay above the existing application interface using a React portal (Meta Open Source, 2026). The overlay component receives the validated specification tree and recursively dispatches each node to its corresponding catalog component based on the node's type. A concrete generated example is shown in Appendix A.3. This dispatch model means that the renderer does not need to know about the content of any particular specification; it only needs to map types to components and pass through validated properties.

Most importantly, ephemeral components are not rendered by storing or injecting AI generated HTML directly in the website's page. The validated JSON in `EphemeralSpec` with its metadata and output is stored when after generation. For example, when the nodes are being rendered, each allowed node types is mapped to its own React component, like `InspectPanel`, `AnchoredTooltip`, or `HighlightRing`. The visible overlay is then made with these interface primitives instead of random components made by the LLM.

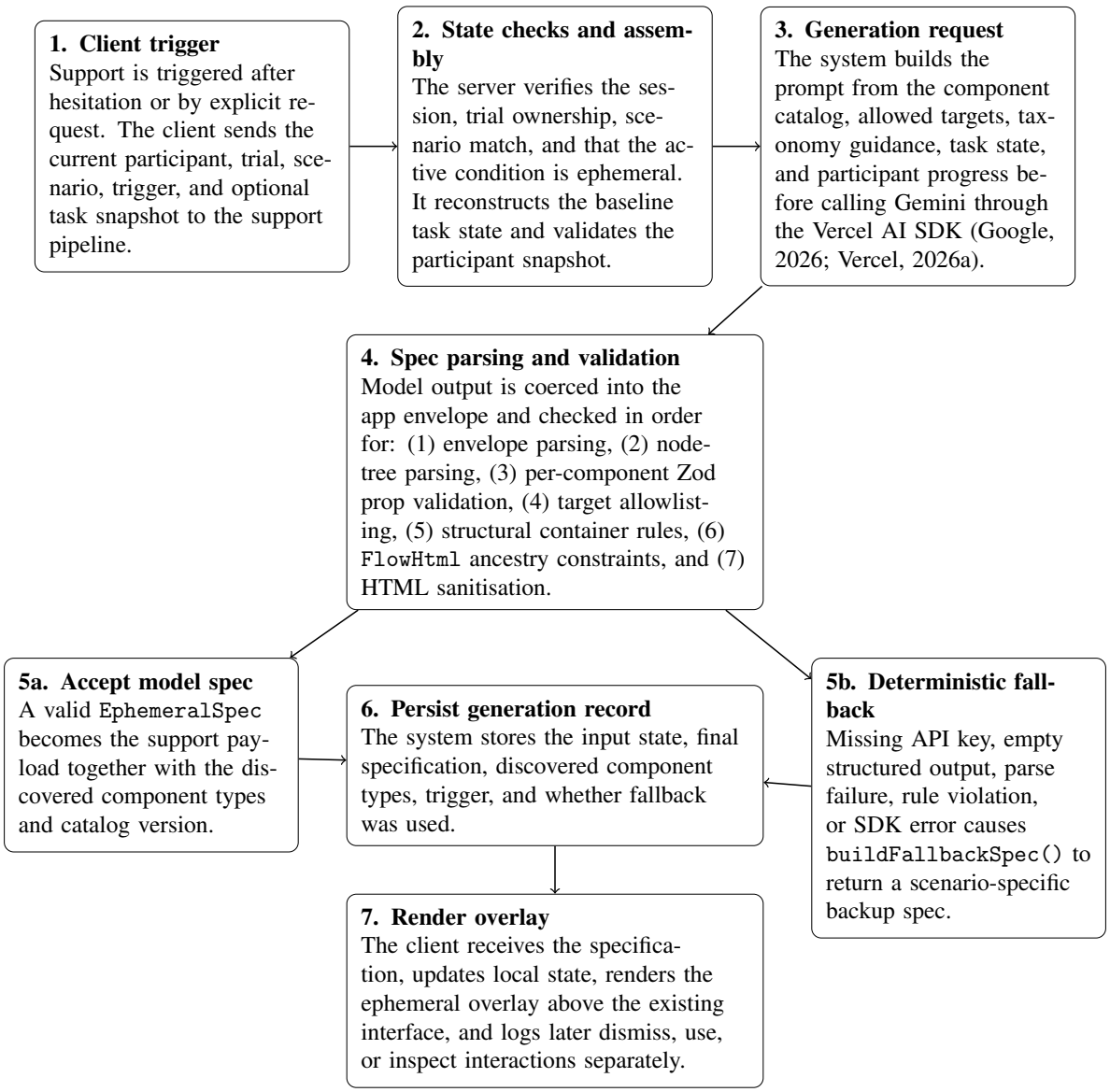


Figure 6: Ephemeral support generation and validation pipeline. Model output passes through seven validation stages before rendering; invalid output triggers a deterministic fallback. Both paths are logged for later analysis.

`HighlightRing`, `AnchoredTooltip`, and `ConnectorLine` reference target elements by looking up DOM elements by identifier. Then a custom hook is used to watch the container and updates the overlay position if layout changes. For example, if any elements were to shift the layout like toggling a sidebar the overlay would update its position as well. Also, the clamping utility ensures that elements remain visible in the viewport even if the target is at the end of the screen.

Overlays respect the lifecycle stage of dismissal defined in methodology. If the specification of metadata says it can be dismissible, it provides commonly a button to let the participant remove the ephemeral support whenever they chosen. Also, these dismissals are logged to understand the difference between support that was used and support that was removed as it was disturbing the user. The overlay does not persist between pages so ephemerality is preserved.

3.6 Instrumentation and Logging

The evaluation within the methodology specifies several behavioural and self-reported measures the participants complete during the trial. The artifact implements the needed instrumentation to capture the relevant data for each measure.

All interactions on the client are sent to a logging API endpoint on the web app that inserts rows into the `trial_events` PostgreSQL table. Each event includes a label with a specific event type and JSON payload of specific details from a given participant. The actual types of events come from a predefined set for recognisable interactions. Also, an interaction count is incremented for whenever these events are recorded.

Importantly, more data is captured for the ephemeral condition than just interaction events. Every time the support generation endpoint is called the full input state with the task state, current participant snapshot, and output specification gets written into the `support_outputs` table. Thus, it's possible to analyse whether the support was actually used, what the system produced and why, and if a fallback was triggered or not due to the limitations of the LLM model.

Regarding the structure of the questionnaire responses from the participants, it is stored as question response pairs. For the specific trial post-trial ratings for intrusiveness, perceived

Table 4: Logged event types and their relationship to evaluation measures. Events marked with an asterisk (*) increment the trial's interaction count.

Event type	When recorded	Evaluation measure
`trial_viewed`*	Participant opens a trial	Interaction count
`slide_reordered`*	A slide is dragged to a new position	Interaction count
`answer_selected`*	An answer is first chosen	Interaction count
`answer_changed`*	An answer is revised	Interaction count
`support_requested`*	Participant explicitly requests support	Support engagement
`support_triggered`*	Support is triggered automatically	Support engagement
`support_shown`*	Support overlay becomes visible	Support engagement
`support_dismissed`*	Participant closes the overlay	Support engagement
`support_used`*	Participant acts on the support	Support engagement
`support_ignored`*	Support is present but not interacted with	Support engagement
`support_inspect_expanded`*	Participant expands an `InspectPanel` detail section	Support engagement

control, and helpfulness are linked. On the other hand, comparative questions are linked to only the participant. By having this structure, it allows for direct joins between the behavioural data and subjective ratings at the trial level.

Finally, to export the results for data analysis there is an export endpoint for participant records, performance metrics, event logs, questionnaire responses and ephemeral UI generation outputs. As a result, providing the full dataset needed to analyse and evaluate the final results described in the methodology.

4 Results

This section reports the outcomes of the comparative user study. All measures follow the analysis plan described in the methodology: behavioural data was extracted from the experiment database, and self-reported ratings were collected through post-trial and post-study questionnaires administered within the same website. Unless stated otherwise, paired comparisons use the Wilcoxon signed-rank test, medians and interquartile ranges (IQR) are reported for continuous and ordinal measures, and statistical significance is assessed at $\alpha = .05$.

4.1 Participants and Data Overview

A total of 48 participants accessed the study website. Of these, 23 completed both task trials and the final questionnaire and are included in the analysis. The remaining participants were excluded because they did not finish both conditions or did not complete the post-study questions. The reported results therefore describe the completer sample rather than all users who initially opened the study website. Table 5 summarises the background characteristics of the included sample.

Table 5: Participant background characteristics ($n = 23$).

Characteristic	Summary
Age range	19 (6); 20 (1); 21 (3); 22 (9); 23 (3); 24 (1)
Web application familiarity (1–7)	Mdn = 6.0, IQR = 6.0–7.0
AI tool familiarity (1–7)	Mdn = 6.0, IQR = 5.0–7.0
Condition order: baseline first	15 (65%)
Condition order: ephemeral first	8 (35%)

4.2 Task Completion Rate

Task completion was defined as submitting the slide deck in the correct canonical order. In the baseline condition, 14 of 23 participants (60.9%) completed the task correctly. In the ephemeral condition, 17 of 23 (73.9%) completed correctly. McNemar's exact test comparing discordant pairs yielded $p = 0.250$.

The ephemeral condition therefore showed a descriptively higher completion rate than the baseline, but the difference was not statistically reliable in the present sample.

4.3 Task Completion Time

Completion time was measured from the moment the participant started the trial to the moment they submitted their answer. Participants who completed the task in the baseline condition had a median time of 35.7 seconds (IQR: 16.4–73.4), compared to 56.1 seconds (IQR: 22.3–85.1) in the ephemeral condition. A Wilcoxon signed-rank test did not yield a statistically significant difference ($W = 125$, $p = 0.709$, $r = 0.094$). Although the ephemeral median was slightly higher, the result does not indicate a reliable difference in efficiency between conditions.

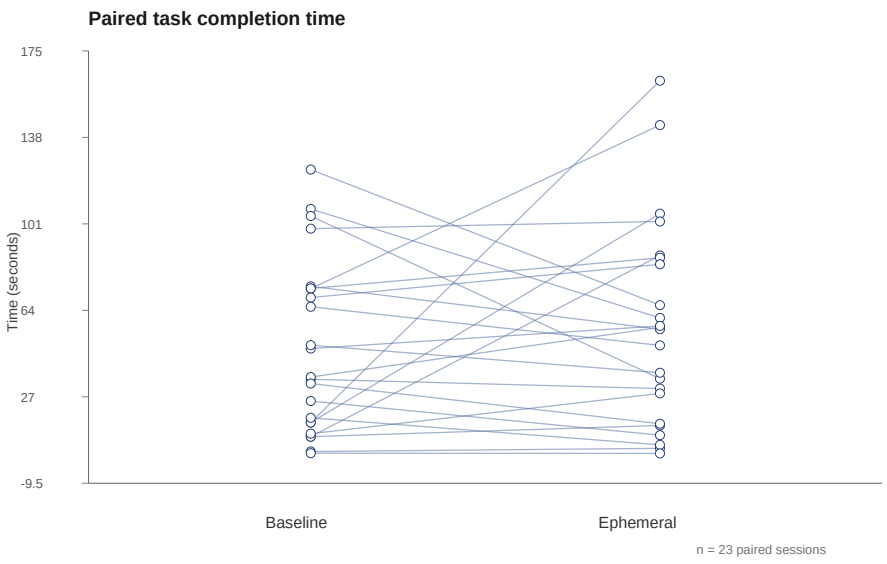


Figure 7: Paired comparison of task completion time (seconds) between the baseline and ephemeral conditions. Each line connects the same participant across conditions.

4.4 Interaction Count

The total number of logged interactions per trial was used as a proxy for task effort. The median interaction count in the baseline condition was 3.0 (IQR: 1.0–4.0), compared to 11.0 (IQR: 6.0–13.0) in the ephemeral condition. A Wilcoxon signed-rank test yielded a statistically significant difference ($W = 0$, $r = 1.000$; $p = <0.001$).

This higher count should be interpreted with caution because the ephemeral condition includes support-related events such as requesting support, support being shown, dismissal, and detail expansion. The increase therefore reflects greater overall interactivity, not only additional effort spent on the underlying slide-reordering task itself.

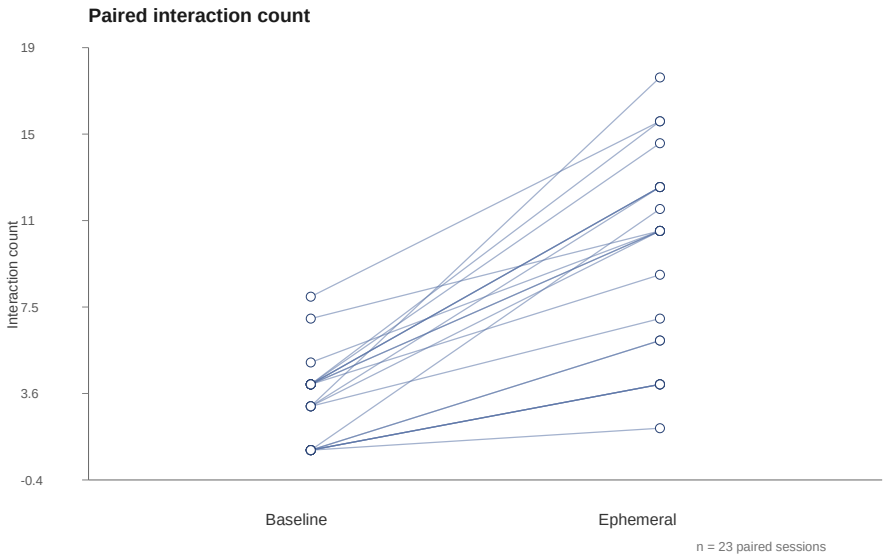


Figure 8: Paired comparison of interaction count between the baseline and ephemeral conditions.

4.5 Subjective Ratings

After each trial, participants rated the interface on three dimensions using a seven-point Likert scale. Table 6 summarises the results.

Table 6: Post-trial subjective ratings (7-point Likert scale, 1 = low, 7 = high).

Measure	Baseline		Ephemeral		Wilcoxon
	Mdn	IQR	Mdn	IQR	p
Helpfulness	5.0	3.0–6.0	6.0	5.0–6.5	0.029
Intrusiveness	3.0	1.5–5.0	5.0	4.0–6.0	0.004
Perceived control	6.0	4.5–7.0	6.0	4.5–6.0	0.579

The ephemeral condition received a higher median helpfulness rating than the baseline (6.0 vs. 5.0), and the Wilcoxon test indicated a statistically significant difference ($p = 0.029$). Thus, participants rated the ephemeral interface as more helpful.

Intrusiveness ratings were higher in the ephemeral condition (median 5.0) than in the baseline condition (median 3.0), and this difference was statistically significant ($p = 0.004$). Participants therefore experienced the temporary support as more noticeable or interruptive than the baseline interface.

Perceived control had the same median in both conditions (6.0 and 6.0), and the paired comparison was not significant ($p = 0.579$). Within the bounds of this study, the ephemeral support did not measurably reduce participants' sense of control over the task.

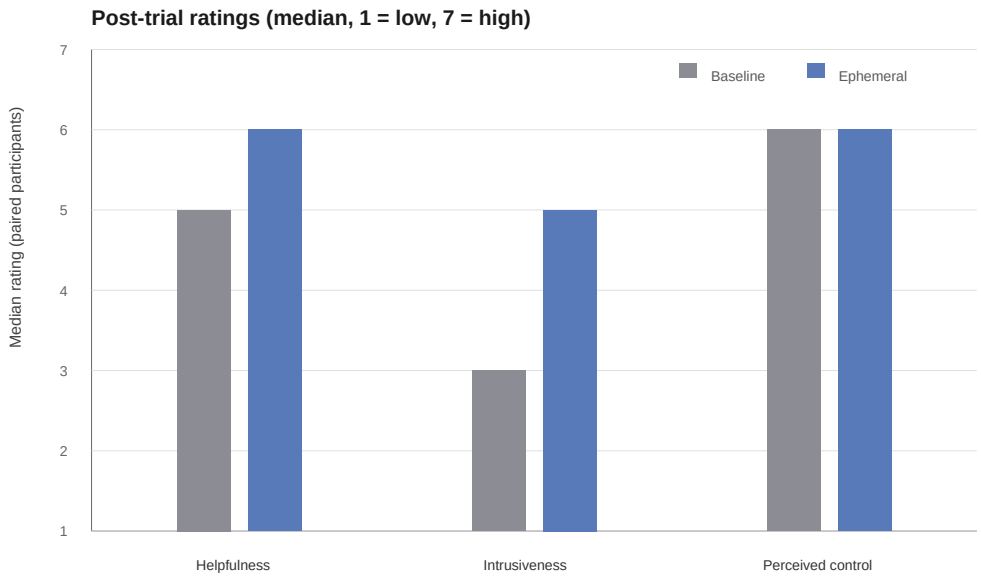


Figure 9: Median post-trial ratings by measure and condition (1 = low, 7 = high). Bars show the median rating across paired participants for each measure.

4.6 Overall Preference

After completing both trials, participants answered four forced-choice comparison questions. Responses were stored as Version A or Version B and mapped back to the actual experimental condition using each participant's counterbalancing assignment. Table 7 summarises the results.

Excluding the 6 participants who expressed no preference, 11 participants preferred the ephemeral condition and 6 preferred the baseline. A binomial test did not show a statistically

Table 7: Final comparative preference questions ($n = 23$).

Question	Baseline	Ephemeral	No preference
Overall preference	6	11	6
More helpful	8	12	3
More intrusive	1	18	4
Real-life preference	9	10	4

significant overall preference difference ($p = 0.332$), indicating that the sample did not meaningfully favour one condition over the other.

4.7 Ephemeral Support Engagement

To interpret the behavioural and subjective results in context, this subsection reports how participants actually interacted with the ephemeral support layer. These measures apply only to the ephemeral condition.

Of the 23 participants who completed the ephemeral trial, support was triggered or explicitly requested in 21 cases (91.3%). At the participant level, 6 participants recorded at least one `support_used` event, 21 recorded at least one `support_dismissed` event, 2 recorded at least one `support_ignored` event, and 3 recorded at least one detail-expansion event. These categories are not mutually exclusive: a participant could, for example, inspect or use support and later dismiss it in the same trial. The support generation pipeline produced 47 specifications in total, and no deterministic fallbacks were recorded (0 of 47, 0.0%).

This engagement pattern indicates that participants were usually exposed to the ephemeral support layer, but many chose not to engage deeply with it once it appeared. The condition-level outcomes should therefore be interpreted in light of partial uptake: the measured effects may reflect both the quality of the support itself and the fact that substantial portions of the offered assistance were dismissed after limited inspection or left unused altogether.

4.8 Summary of Findings

The ephemeral condition produced a higher descriptive task completion rate than the baseline, but the difference was not statistically significant. Completion time also did not differ signifi-

cantly, with the ephemeral condition showing a slightly higher median time. Interaction count was substantially higher in the ephemeral condition, indicating greater overall interface activity during the trial. Subjective ratings suggested that the ephemeral interface was significantly more helpful, significantly more intrusive, and no different in perceived control. Final preference responses did not reveal a statistically reliable overall preference for either condition. Engagement data shows that support was usually surfaced, but sustained use was limited, which provides important context for interpreting the mixed behavioural and subjective outcomes.

Table 8 provides an overview of all primary outcome measures.

Table 8: Summary of primary outcome measures across conditions.

Measure	Baseline	Ephemeral	p
Completion rate	60.9%	73.9%	0.250
Completion time (Mdn, s)	35.7	56.1	0.709
Interaction count (Mdn)	3.0	11.0	$<$0.001
Helpfulness (Mdn)	5.0	6.0	0.029
Intrusiveness (Mdn)	3.0	5.0	0.004
Perceived control (Mdn)	6.0	6.0	0.579
Overall preference	6 chose baseline; 11 chose ephemeral; 6 no preference		0.332

5 Discussion

This study evaluated whether ephemeral, task-scoped generative UI can improve a bounded productivity task. The final results show that it's a mixed answer. Relative to the baseline, the ephemeral version produced a higher descriptive completion rate, higher helpfulness rating, and stable control. However, completion time or overall performance did not improve. Furthermore, it significantly increased the interaction count and perceived intrusiveness. These findings suggest that ephemeral support can be integrated into modern day applications without disrupting the user experience and without neglecting user control. Though, this experiment as a whole was not able to deliver a clear performance gain without introducing more friction to the user.

5.1 Interpretation and Relation to Prior Work

This study's focus is different to most of the recent discussion around generative UI. Prior work emphasised on adaptability, multimodality, and the capability of these AI systems to tailor the interaction of users across contexts (Bieniek et al., 2024). Prior literature explains why static interfaces are insufficient in environments where user needs change rapidly. Additionally, it shows that the same adaptability that makes generative UI interesting can also threaten predictability, continuity, and user control. This study addresses that dilemma by going into a narrower alternative. Instead of replacing the entire interface dynamically, the artifact constrains the generative support to a temporary and task-scoped layer to avoid restructuring the entire user workflow. The results suggest this approach could provide benefits for the confidence of the user rather pure efficiency. Participants rated the ephemeral condition as significantly more helpful, but they did not complete the task faster and no reliable increase in success.

The results of this capstone support the claims that were deduced in the BISCUIT study (Cheng et al., 2024). It showed that ephemeral UIs can act as a intermediary between user intent and LLM-generated output in computational notebooks. In that study, the temporary interface made it easier for users to understand the code that was generated, look at other options, and lessen the work of prompt engineering. This work builds on that idea but moves it to a new

area. Instead of helping novice programmers who might rely completely on AI to help with the data analysis, it showed provides higher productivity gains for experienced programmers and preserve their orientation in an existing application. The weaker and more mixed effects seen here suggest that the current implementation was not as interactive as BISCUIT's notebook scaffolds. In this case, all participants encountered the support but only a fraction actually used it. Thus, lightweight overlays may not yield significant benefits unless they can provide immediate value to warrant that attention.

The discussion also connects to the earlier concept of ephemeral user interfaces proposed by Döring et al., who argue that intentionally temporary interface elements can reduce overload by limiting persistent accumulation and by presenting information only for the period in which it is relevant (Döring et al., 2013). Their work was grounded primarily in material and interaction-design perspectives rather than in web-based productivity systems, yet its core logic remains relevant here. The present thesis translates that logic into a digital software context by treating ephemerality not as a material property, but as a design constraint on interface behaviour. The subjective results highlight both sides of that design logic: perceived control remained stable, which supports the claim that the support stayed sufficiently bounded and did not take over the workflow, but intrusiveness increased significantly, showing that temporary appearance alone does not guarantee a low-disruption experience. In other words, bounded scope seems achievable, but trigger timing, presentation, and the immediate usefulness of the support remain critical if ephemerality is to feel genuinely unobtrusive.

All in all, these comparisons demonstrate that it's not needed to make a final verdict on the whether generative UI has a definitive place in productivity tools, but that it has a place in the middle in the literature. While static interfaces preserve continuity it struggles with flexibility and task specific assistance, fully generated interfaces provide unlimited flexibility with a risk of loss of user orientation and trust. The framework in this capstone proposes that ephemeral, task-scoped ui can offer a compromise between the two extremes. It does so by constraining the adaptive assistance through a boundaries of trigger conditions, scope, continuity, user control, and dismissal. The results suggest that the ephemeral support can be added without loss of perceived control. Nonetheless, it does indicate that the current intervention was too narrow,

intrusive, and weakly taken up to show stronger performance benefits.

The findings suggests that ephemeral support does not suit all productivity tasks in general, but in constrained specific subtasks. These tasks include manually adjusting variables for instant feedback like in data analysis or design exploration. Temporary support can add value while still leaving the underlying application stable and familiar. Though, in regulated industries these actions could be restricted on when the ephemeral support would be accepted, especially in areas such as compliance or healthcare. However, the current results suggest that this support should remain optional and immediately useful. Otherwise the cost of interruption can outweigh the benefit of assistance.

5.2 Technical Iterative Refinement Contexts

Although the thesis does not directly evaluate design tooling or data-analysis software, the pattern of results suggests that such contexts are worth treating as plausible next-step evaluation targets rather than as confirmed application domains. The benefit signal in the current study was not faster task execution, but higher perceived helpfulness under stable perceived control.

Which means that ephemeral UI can be used for refinement oriented tasks instead of just benefiting raw efficiency. Thus, more technical workflows that involve constant tweaking of parameters in need of immediate feedback could be a stronger fit for future testing than workflows that depend on uninterrupted speed.

For example, it could be for designers adjusting the layout of a dashboard on the actual web app getting real time feedback instead of a static design canvas. Or even a data analyst who is tuning the parameters of a logistic regression model to see if it's overfitted. In both cases, the user is making several iterative changes in a static interface, fitting well with the framework's focus on limited scope, continuity, and user control. However, this is merely a hypothesis based on the data inferred from the artifact in the slide refinement scenario. Hence, it is possible to justify these technical tweaking contexts as next steps for the evaluation but not hard evidence that ephemeral UI works the best in these scenarios.

5.3 Limitations

The main limitation of this study was the scope of the actual evaluation. The artifact had only one task of fixing the order of a given slide deck based on some specific context rather than a full range of different support roles that were proposed in the framework. This choice was appropriate for the capstone as it reduced participant fatigue while completing the survey. A design with three scenarios across two interface conditions would have required six task trials per participant, which would likely have increased drop-off, reduced engagement, and weakened the quality of responses. However, this decision means that the findings cannot yet be generalised across multiple task types, domains, or support roles such as interpretive support or task-execution support.

The study is also limited by the breadth of interactions represented in the artifact. Some richer or more open-ended forms of support, such as more advanced writing assistance or more expressive temporary interaction patterns, were not retained in the final evaluation design. This means that the study primarily evaluates a bounded form of ephemeral support in a structured workflow rather than the broader design space of adaptive generative interfaces. As a result, the conclusions are strongest for short, goal-directed refinement tasks and should not be treated as evidence that the same interaction logic would perform equally well in longer writing workflows, highly creative tasks, or more complex multi-step activities.

Other limitations are sampling, device conditions, and session duration. It was important to avoid gathering too much personal data of participants like knowing more about their background and habits for the sake of anonymity. However, it impeded the extent to how ephemeral UI could be explored more using the context of the user personally. Furthermore, some participants accessed the study on smaller screen sizes like their phones. The artifact was not optimised for phone use so it introduced additional friction unrelated to the experiment so it could have skewed completion time, interaction effort, perceived intrusiveness, or overall preference. Furthermore, participants only interacted with the study for a short amount of time. Thus, conclusions could only be made for immediate task performance and not for longer term effects such as habituation of the ephemeral assistance once novelty worn off.

Furthermore not all participants completed the entire study. 48 participants accessed the web app and only 23 completed both trials and the final questionnaire. Thus, the analysed sample size likely overpresents participants willing to completely the full process. Results should be interpreted as evidence about those who actually completed it rather than everyone who initially encountered the study.

Another limitation is that framework was only tested on one specific implementation rather than several. The positive or negative outcomes then reflect not only the framework but specific design decision in the prototype. For example, these include how support was communicated, when it was triggered, and how the interaction options were shown. Subject measures such as helpfulness, perceived control, and intrusiveness rely on the participant responding truthfully. Thus, all the questionnaire responses are shaped by the individual's own perceived interpretation. These limitations don't invalidate all the findings, but emphasise that the results should be evaluated under the operationalisation of the framework rather than a definitive test of ephemeral support in the abstract.

The paired comparison analysis is limited by size of the data as well as structure from participants interacting with ephemeral UI in the host application. The order of whether to show ephemeral or traditional version of the UI was randomised across participants to help reduce systematic ordering bias. However, the final sample size is too small to support the claims of the order effects. Also, the engagement metrics from the temporary support UI show multiple event types instead of just one participant state. Henceforth, the data on the frequency of support being shown, interacted, or dismissed can only be seen as informative as opposed to a complete behavioural taxonomy.

Another limitation is that the study evaluates only visual representation of ephemeral support. The artifact uses primarily an overlay with deterministic properties with some to keep the interface consistent. However, that design choice reflects only one way of instantiating ephemerality. Temporary support could be added directly in the applications existing layout instead of on top of it or even by extending and existing component without needing to introduce a distinct overlay. This means that the present results should not be interpreted as evidence about all possible forms of ephemeral UI, but specifically about an overlay-based implementa-

tion. Thus the cost of intrusiveness that came from the results could be a consequence of not further exploring different ways of fundamentally representing ephemerality.

5.4 Robustness

Nonetheless, several parts of how the research design was conducted strengthen the robustness of the study. Both interface variations evaluated the same task with the same success criteria for a fair comparison between baseline and ephemeral versions. The within subjects design reduced any chance of a fake difference as participants encountered both variations in randomised trial order. The web app also automated event logging with the post-trial questionnaires to make it possible to compare the behavioural outcomes and subjective experience. The framework rules about scope, lifecycle, and user control limited the variation from the ephemeral implementation. Hence, the ability to analyse and compare between sessions was preserved.

One unexpected finding was that the behavioural and subjective measures do not align. The ephemeral versions in the study was considered significantly more helpful with high completion rate, but resulted in no significant improvement in finishing the task while being more intrusive. This suggests that participants recognised the intent or potential value of the ephemeral support. However, it did not feel seamless enough to outweigh its interaction cost for such a short task. This contradiction is important as it implies that in future work less focus should be put on adding more support and more focus on making that support feel faster, lightweight, and engaging to use. Taken together, these limitations and robustness measures suggest that the study should be interpreted as a focused but credible first evaluation of ephemeral, task-scoped support in a stable-interface software setting. The results are sufficient to discuss the promise of the framework in a controlled setting, but they do not establish broad generalisability across contexts. Section 6 sets out concrete directions for extending scenarios, evaluation methods, and the design space of ephemeral components.

6 Future Work

The present evaluation is intentionally narrow: one scenario, one operationalisation of the framework, and a remote, self-paced procedure. The directions below follow from that scope and from what the artifact already suggests about where ephemeral, task-scoped support may be most informative next.

6.1 Broader scenarios and user populations

The study used a single presentation-refinement workflow. Ephemeral support is unlikely to matter equally in every domain. Prior work like with BISCUIT (Cheng et al., 2024) suggest that ephemeral interface scaffolds work well with computational notebooks for data science exploration where they can steer intermediate output. Thus, it provides a useful contrast to this thesis where it focused on specific technical workflow, whereas this projects focuses if ephemeral support can add value in conventional interfaces for a larger user demographic. The current results show that less focus should be made on broad domains and more on the workflow properties of interruption tolerance, consequence of error, and whether the assisted action is a bounded local refinement rather than a high-stakes commitment. The most interesting scenarios to test next would be technical refinement tasks through several iterations. For example, when a designer wants to tweak the layout or hierarchy of their designs or a data scientist adjusting the parameters of a machine learning model they both require real time feedback from their actions. The same reasoning indicates significant limitations regarding the appropriateness of ephemeral support. Within heavily regulated areas like finance, law, or healthcare ephemeral generative UI may not be an ideal use case as those scenarios. This is as they require stricter predictability, traceability, and accountability than low risk situations like adding tasks to a todo list, even if it's proven useful for refinement tasks. This possibility also aligns with the claim that adaptive support could be the most valuable when traditional interfaces struggle to accommodate varying user needs across different contexts (Bieniek et al., 2024; Tambi, 2020). The next step is to evaluate the framework across several workflows that differ in interruption tolerance, consequence of error, task significance, technical complexity, familiarity with the

application, and degree of regulatory constraint. It would clarify situations when ephemeral support is genuinely helpful for end users or if it gives irrelevant guidance for a clear user workflow.

6.2 Deeper evaluation methods

Increasing the number of completions would make the statistics more reliable, but it wouldn't explain why participants acted the way they did or how they understood the support. Another direction would be to do in depth evaluation in focus groups where a researcher can observe the misunderstanding of triggers and emotional reactions of the participants that questionnaires can only partially capture. Using both controlled experiments and focus groups could link specific design choices to real friction and also to how intrusive or controllable the support felt.

6.3 Richer ephemeral component catalogues

The evaluation artifact deliberately used a constrained set of ephemeral UI components so that conditions remained comparable and the study remained feasible within the capstone timeline. The underlying framework could support a wider catalogue: more varied layouts, stronger personalisation of content and tone based on the user's profile, role, or prior behaviour, and more expressive temporary interaction patterns than were deployed here. Exploring that wider design space would also connect more directly to the broader literature, which frames generative interfaces in terms of adaptive, context-sensitive interaction structures while also warning that flexibility can come at the cost of predictability and user control (Bieniek et al., 2024). At the same time, the foundational logic of ephemerality suggests that temporary interface elements should appear only when relevant and should recede without leaving persistent clutter (Döring et al., 2013). Expanding the catalogue therefore raises its own research questions about how to preserve predictability and user agency when surface behaviour varies more widely, and how to evaluate more personalised variants without confounding condition effects with implementation novelty. Future work could treat catalogue design as a first-class object of study: progressive disclosure of complexity, user-tunable boundaries for ephemerality, and systematic comparison of a small number of high-diversity implementations against a stable baseline.

7 Conclusion

This thesis asked how ephemeral, task-scoped generative UI components can be designed to improve existing applications without disrupting the user experience. To address that question, it developed a conceptual framework for ephemeral support, implemented the framework in an experimental web artifact, and evaluated the artifact against a persistent baseline in a within-subjects user study. The evaluation provides a qualified answer to that question: ephemeral support can be introduced in a bounded way that preserves users' sense of control, but the implementation tested here did not produce clear performance gains without also introducing additional intrusiveness.

The temporary variation had a slightly higher task completion rate, but the difference wasn't big enough to be statistically significant. It also had a much higher perceived helpfulness rating. Completion time did not get better. However, interaction count increased substantially and perceived intrusiveness increased significantly, while the perceived control remained stable while participants were using the ephemeral variation. Though, for preference there was no clear favour in either condition. The results suggest that the ephemeral layer in this prototype did not help faster task execution, but a clearer sense of perceived support with noticeable interaction cost.

The capstone's main contribution is that it brings ephemerality from a theory to a tangible proof of concept. It does so by using a component model specification, generative AI pipeline, and constrained ephemeral overlay to preserve the original interfaces of applications. It did this through an interactive survey artifact for empirical evaluation.

As a result, ephemeral task-scoped generative UI is a promising design direction for software design. However, it is more complex than just creating temporary support for some user. For applications with better user experience, ephemeral UI needs to be shown at appropriate times, easy to understand, and useful enough so users don't ignore it. At the moment, it's more suited to extremely targeted tasks within a larger workflow. It works particularly well in helping with contextual guidance when interruptions can be tolerated. The framework proposed in the thesis defines those boundaries, while showing promise and current limits for ephemeral UI.

Processes that need iterative changes such as exploring different designs or parameter tuning with instant feedback are plausible next steps for this study. Future work should test these richer scenarios with a broader set of participants and greater variety in temporary UI behaviours to determine when ephemeral generative assistance will become meaningful to society.

In conclusion, I believe what has been built for this project expands beyond a simple research paper as it provides an actual artifact that can be used and extended in real production apps. I believe that ephemeral UI is a promising direction for design and will continue to change as the space of generative UI changes. The most difficult challenge to balance is understanding when and how an ephemeral component should be displayed to a user while being constrained to the application's design system. I think this work matters beyond the experiment as it marks early steps in a broader shift across the design and software industry. The way we think about building software products may change as ephemeral UI creates flexibility in user journeys that was not previously possible while still aiming to give people better experiences.

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A Appendix

This appendix documents the key analysis steps used to transform the study database into the statistics, figures, and thesis-ready result values reported in Section 4. Because the examiner cannot assume access to the full repository, the appendix includes selected code excerpts that demonstrate the core logic of participant selection, statistical testing, figure generation, and LaTeX value generation. The excerpts are representative rather than exhaustive: they are intended to show how the analysis was implemented, not to reproduce every utility function in full.

A.1 Analysis Pipeline Overview

The analysis workflow was implemented as a small Bun-based pipeline. Rather than copying values manually into the thesis, the pipeline exported study data from PostgreSQL, computed the required descriptive and inferential statistics, generated the figure PDFs used in the results chapter, and wrote the LaTeX macro file that supplied numeric values to the manuscript. This structure reduced transcription risk and kept the reported values tied to executable analysis code.

The pipeline can be summarised in four stages:

1. export the required study data to intermediate CSV files,
2. compute the paired statistical tests used in the results chapter,
3. generate the PDF figures included in the results chapter, and
4. regenerate the LaTeX macros that populate reported values throughout Section 4.

This means that the values appearing in the results chapter were derived from the exported data rather than entered by hand. If the underlying database contents changed, rerunning the pipeline updated both the figures and the numeric placeholders used in the LaTeX source.

A.2 Representative Analysis Listings

The following listings were selected because they show the most thesis-relevant parts of the implementation: the rule used to determine the analysed sample, the paired non-parametric statistical test used for several outcomes, the generation of the main numeric figures, and the production of thesis-ready LaTeX commands.

A.2.1 Participant Inclusion Logic

Listing 1 shows the reusable SQL filter used throughout the analysis. A participant was included only if they completed both task trials and answered all four final comparison questions. This ensured that all reported measures were computed from a consistent completer sample.

A.2.2 Paired Statistical Testing

Listing 2 shows the central implementation of the Wilcoxon signed-rank test used for paired completion time, interaction count, and post-trial Likert outcomes. The function removes zero differences, ranks the remaining paired differences by magnitude, computes the smaller rank sum W, and then uses an exact p-value for small samples or a normal approximation for larger ones. This matched the within-subjects design and the small paired sample available in the study.

A.2.3 Figure Generation

Listing 3 shows the core drawing loop used to produce the paired numeric plots for completion time and interaction count. Each participant contributes one baseline value and one ephemeral value, and the script draws a line connecting the two observations so that the within-participant change is visible directly in the figure.

A.2.4 Thesis-Ready Value Generation

Listing 4 shows how the final numerical outputs were converted into LaTeX commands. Instead of entering p-values, medians, and percentages manually into the thesis text, the script generated commands such as `\GenTimeP` and `\GenCompPctBaseline`. The results chapter

then imported these commands directly, ensuring that the prose, tables, and summary values all used the same generated source.

A.3 Local Generation Trace

In addition to the aggregate study dataset analysed in Section 4, I captured a local end-to-end validation trace from the implemented support pipeline itself. This trace was taken from the real `/api/support` route rather than the development-only playground endpoint, so it exercised the same request handling, prompt construction, validation stages, PostgreSQL persistence, and client-facing response path used during the study. The selected example below is the locally persisted record with `support_output_id = 103`, captured on 2026-04-04 during a hesitation-triggered `slides-outline-refine` trial.

Listing 5: Condensed excerpt of a locally captured support-generation record.

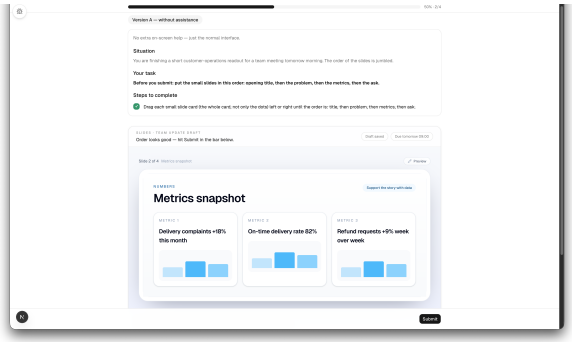

```
{
  "date": "2026-04-04",
  "route": "/api/support",
  "scenarioId": "slides-outline-refine",
  "participantId": "192571eb-71f5-4e69-9808-b1dedb71bdf6",
  "trialId": "f8523735-82a2-4926-9e7a-26a78eb13a19",
  "summary": {
    "successfulRuns": 3,
    "supportOutputIds": [101, 102, 103],
    "modelName": "gemini-2.0-flash",
    "allUsedFallback": false,
    "allProviderAttempted": true,
    "triggers": ["hesitation"],
    "observedComponentTypeSets": [
      ["Stack", "StepRail", "InspectPanel", "ConnectorLine"],
      ["Stack", "StepRail", "InspectPanel"],
      ["Stack", "StepRail", "InspectPanel", "ConnectorLine", "AnchoredTooltip"]
    ]
  },
  "selectedExample": {
    "requestId": "507d8f6e-525b-4478-8870-e9ac1be8e625",
    "supportOutputId": 103,
    "supportOutputCreatedAt": "2026-04-04T21:21:21.820Z",
    "trigger": "hesitation",
    "generation": {
      "modelName": "gemini-2.0-flash",
      "usedFallback": false,
      "fallbackReason": null,
      "providerAttempted": true,
      "catalogVersion": "catalog-v3",
      "componentTypes": [
        "Stack",
        "StepRail",
        "InspectPanel",
        "ConnectorLine",
```

```
      "AnchoredTooltip"
    ]
  },
  "inputExcerpt": {
    "participantSnapshot": {
      "scenarioId": "slides-outline-refine",
      "slideOrder": [
        "slide-b-meeting",
        "slide-b-policy",
        "slide-b-proof",
        "slide-b-vote"
      ]
    }
  },
  "persistedOutput": {
    "catalogVersion": "catalog-v3",
    "spec": {
      "version": 1,
      "root": {
        "type": "Stack",
        "props": {},
        "children": [
          {
            "type": "StepRail",
            "props": {
              "targetIds": [
                "slide-b-meeting",
                "slide-b-policy",
                "slide-b-proof",
                "slide-b-vote"
              ]
            }
          },
          {
            "type": "InspectPanel",
            "props": {
              "targetId": "deck-context-bar",
              "title": "Suggested Flow",
              "summary": "Try this order: meeting context -> relevant policy -> proof the society qualifies -> ballot-ready asks. Frame the
              ↪  vote first, then explain the rules, show why this society qualifies, and end with clear requests.",
              "details": [
                "Opening with context sets the stage.",
                "Highlighting policy grounds the story in the union's rules.",
                "Evidence builds confidence in the society's track record.",
                "Ending with concrete requests ensures a focused discussion."
              ]
            }
          },
          {
            "type": "ConnectorLine",
            "props": {
              "fromTargetId": "slide-b-policy",
              "toTargetId": "slide-b-proof"
            }
          },
          {
            "type": "AnchoredTooltip",
            "props": {
              "targetId": "slide-b-policy",
              "body": "Policy before proof: explain the rules before showing how the society meets them.",
              "placement": "top"
```

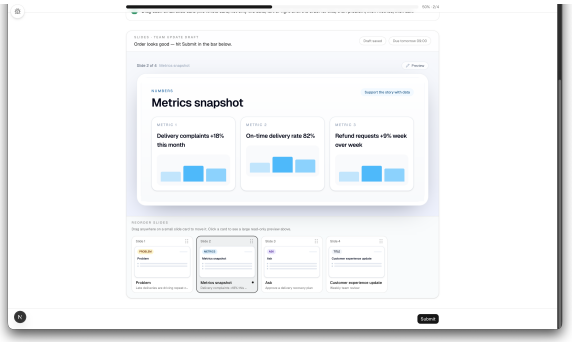
```
            }
          }
        ]
      },
      "meta": {
        "dismissible": true
      }
    }
  }
 }
}
```


A.4 Participant Flow Screenshots

Figures 10–14 provide a chronological walkthrough of the participant-facing flow captured from the implemented artifact. These appendix figures complement the main-text interface screenshots by showing the sequence of states the participant moved through across the baseline trial, the ephemeral trial, and the final comparative questionnaire. One intervening transition capture that contained no visible interface content was omitted.



(a) Baseline trial with the plain interface visible and no temporary support.



(b) Baseline task interaction while the participant reorders the slide strip.

Figure 10: Participant-flow walkthrough, part 1: baseline-condition task screens.

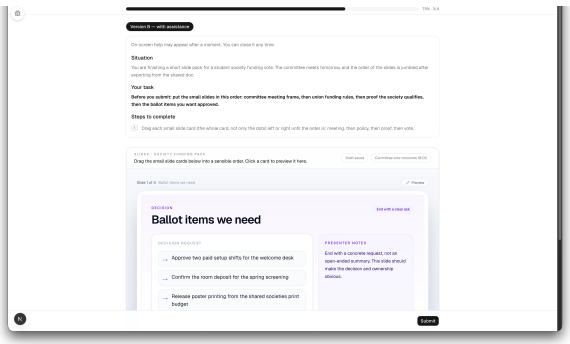
A.5 Interpretive Note

These listings are sufficient to show the main route from raw study records to reported results: first the valid sample was defined consistently, then paired statistical tests were run, then publication-quality figures were generated, and finally the resulting numerical outputs were written into LaTeX commands consumed by the manuscript itself. This is the core evidence needed to demonstrate that the quantitative results chapter was generated through a

```sql
WITH valid_participants AS (
  SELECT p.id, p.age_range, p.occupation,
         p.web_app_familiarity, p.ai_tool_familiarity,
         p.baseline_is_version_a
  FROM participants p
  WHERE (SELECT count(*) FROM trials t
         WHERE t.participant_id = p.id AND t.completed = true) = 2
    AND (SELECT count(DISTINCT question_key)
         FROM questionnaire_responses qr
         WHERE qr.participant_id = p.id
           AND qr.trial_id IS NULL
           AND qr.question_key IN (
             'final_preference',
             'final_helpfulness',
             'final_intrusiveness',
             'final_real_life'
           )) = 4
)
SELECT count(*) AS valid_completed
FROM valid_participants;
```

Listing 1: SQL logic used to define the valid participant sample.

(a) Baseline post-trial questionnaire immediately after task completion.



(b) Start of the ephemeral-condition trial, which warns that temporary help may appear.

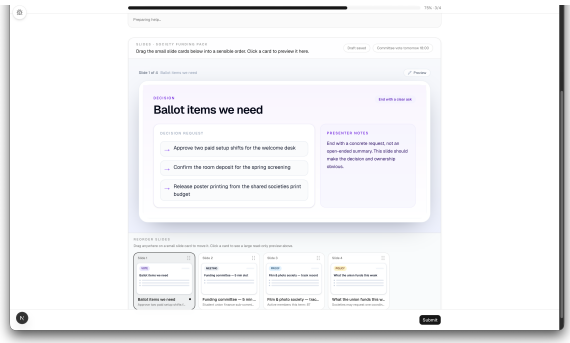
Figure 11: Participant-flow walkthrough, part 2: transition from the baseline trial to the ephemeral trial.

```
function wilcoxonSignedRank(
  baseline: number[],
  ephemeral: number[],
): { W: number; p: number; r: number; n: number } {
  const diffs: number[] = [];
  for (let i = 0; i < baseline.length; i++) {
    const d = ephemeral[i]! - baseline[i]!;
    if (d !== 0) diffs.push(d);
  }

  const n = diffs.length;
  if (n === 0) return { W: 0, p: 1, r: 0, n: 0 };

  const absDiffs = diffs.map((d) => ({ abs: Math.abs(d), sign: Math.sign(d)
  ↪  }));
  absDiffs.sort((a, b) => a.abs - b.abs);

  const ranks: number[] = [];
  let i = 0;
  while (i < absDiffs.length) {
    let j = i;
    while (j < absDiffs.length && absDiffs[j]!.abs === absDiffs[i]!.abs) j++;
    const avgRank = (i + 1 + j) / 2;
    for (let k = i; k < j; k++) ranks.push(avgRank);
    i = j;
  }

  let wPlus = 0;
  let wMinus = 0;
  for (let k = 0; k < n; k++) {
    if (absDiffs[k]!.sign > 0) wPlus += ranks[k]!;
    else wMinus += ranks[k]!;
  }

  const W = Math.min(wPlus, wMinus);
  const maxW = (n * (n + 1)) / 2;
  const p = n <= 25 ? wilcoxonExactP(W, n) : wilcoxonApproxP(W, n);
  const r = 1 - (2 * W) / maxW;

  return { W, p: Math.min(p, 1), r: Math.abs(r), n };
}
```

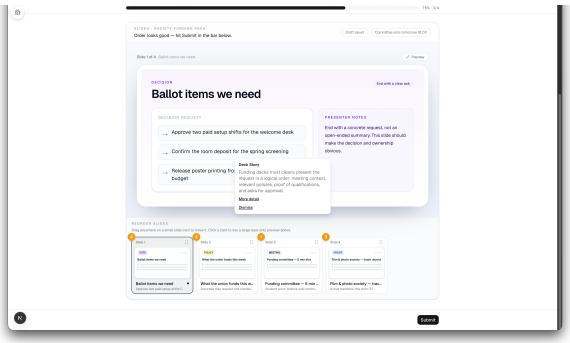
Listing 2: Core Wilcoxon signed-rank implementation used for paired outcomes.

```
const baselineVals = rows.map((r) => Number(r[opts.baselineKey]));
const ephemeralVals = rows.map((r) => Number(r[opts.ephemeralKey]));

for (let i = 0; i < rows.length; i++) {
  const b = baselineVals[i]!;
  const e = ephemeralVals[i]!;
  if (Number.isNaN(b) || Number.isNaN(e)) continue;
  const y1 = yToPdf(b, minY, maxY, chartBottom, chartTop);
  const y2 = yToPdf(e, minY, maxY, chartBottom, chartTop);
  page.drawLine({
    start: { x: xLeft, y: y1 },
    end: { x: xRight, y: y2 },
    thickness: 0.8,
    color: lineColor,
    opacity: 0.55,
  });
}

for (let i = 0; i < rows.length; i++) {
  const b = baselineVals[i]!;
  const e = ephemeralVals[i]!;
  if (Number.isNaN(b) || Number.isNaN(e)) continue;
  const y1 = yToPdf(b, minY, maxY, chartBottom, chartTop);
  const y2 = yToPdf(e, minY, maxY, chartBottom, chartTop);
  page.drawCircle({ x: xLeft, y: y1, size: 2.8, borderColor: pointColor });
  page.drawCircle({ x: xRight, y: y2, size: 2.8, borderColor: pointColor });
}
```

Listing 3: Core paired-plot drawing logic used for numeric result figures.



(a) Ephemeral-condition task before the support overlay expands.

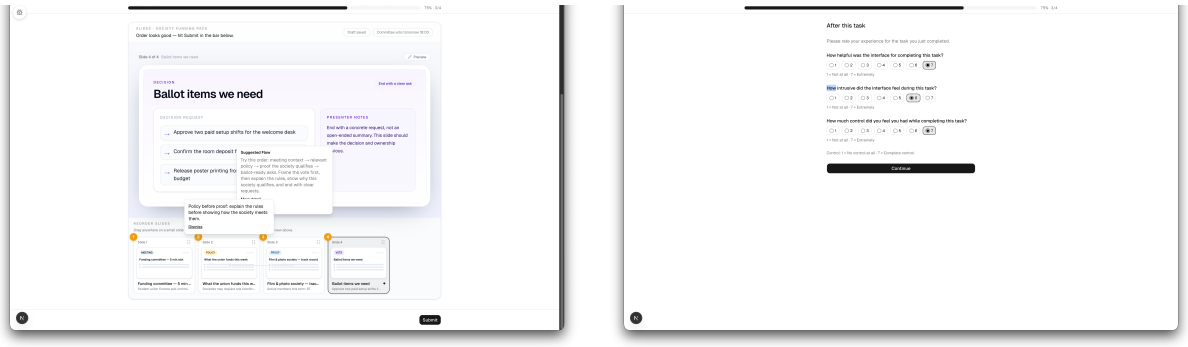


(b) Generated support visible as an anchored explanation and ordered markers over the slide strip.

Figure 12: Participant-flow walkthrough, part 3: ephemeral assistance becoming visible during the task.

```
const lines: string[] = [
  "% AUTO-GENERATED by analysis/generate-results-tex.ts -- do not edit by
   ↪  hand.",
  "% Regenerate: bun run gen-tex",
  "",
];

cmd(lines, "GenNTotal", String(s.overview.totalAccessed));
cmd(lines, "GenNValid", String(v));
cmd(lines, "GenCompPctBaseline", s.completion.pctBaseline.toFixed(1));
cmd(lines, "GenCompPctEphemeral", s.completion.pctEphemeral.toFixed(1));
cmd(lines, "GenCompMcNemarP", fmtP(s.completion.mcnemarP));

const tw = s.time.wilcoxon;
cmd(lines, "GenTimeBaselineMdn", safeFixed(s.time.baselineMdn, 1));
cmd(lines, "GenTimeEphemeralMdn", safeFixed(s.time.ephemeralMdn, 1));
cmd(lines, "GenTimeP", fmtP(tw?.p ?? null));

writeFileSync(OUT_TEX, lines.join("\n") + "\n", "utf-8");
```

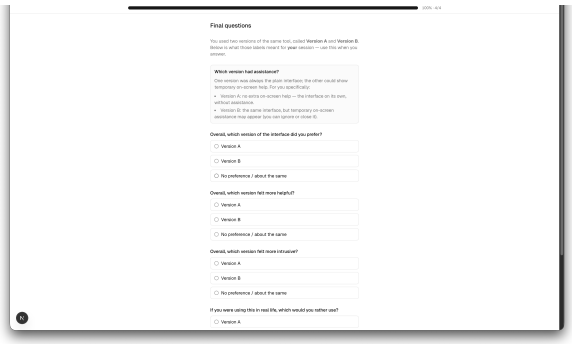
Listing 4: Generation of LaTeX commands used throughout the results chapter.



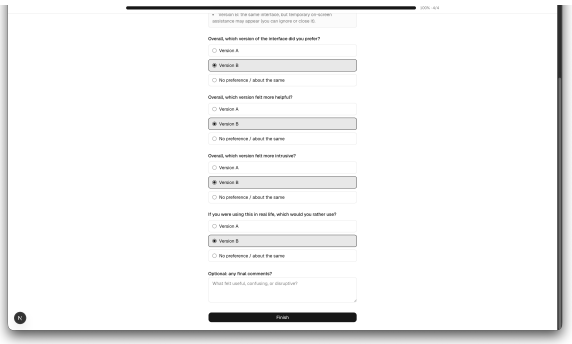
(a) Ephemeral overlay aligned to the current slide and reorder strip.

(b) Expanded support details combining an inspect panel, tooltip, and ordering cues.

Figure 13: Participant-flow walkthrough, part 4: richer support states within the ephemeral trial.



(a) Beginning of the final comparative questionnaire after both trials are complete.



(b) Completed final questionnaire with comparative judgments recorded.

Figure 14: Participant-flow walkthrough, part 5: final comparison questions at the end of the session.

reproducible analysis workflow rather than through manual transcription.